package RAD

imports

DressingService

PlatformOp
PlatformSupport
ISupportCall
IDressingOp
temp: int
retry: boolean, completed: boolean
assent: boolean
IDressing
IRoomTemp
ISupport

DressingComplete
DressingAbandoned
RetryAgreed
assentToSupportCalls: boolean
roomTemperature: int
DressingStarted
SupportCalled

Idle
DressingStarted
[completed]
Dressing
entry roomTemperature?temp <{0}
during Dress(temp)
exit Clear(); DressingComplete <{0}
[retry]/RetryAgreed <{0}
DressingAbandoned
Abandoned
during Retry()
exit Clear()
[not retry/\sinceEntry(Abandoned)>=120]/CallSupportDecision()

Clear()
IComplete
IRetry
/completed = true; retry = false

SupportCalled
MonitoringService
ICurtainOp
PlatformSupport
PlatformOp
ISupportCall
underDressed: boolean
assent: boolean
distress: Distress
ICurtains
IUserParam
IUserFall
ISupport

UserFallen
assentToSupportCalls: boolean
CurtainOpenRqt
CurtainsOpened
RefuseRequest
userUnderDressed: boolean
userDistressed: Distress

Idle
UserFallen/CallSupportDecision()
CurtainOpenRqt/userUnderDressed?
underDressed <{0}; userDistressed?distress <{0}
OpenCurtains
entry OpenCurtains()
[not (underDressed/\distress!=Distress::high)]
[underDressed/\distress!=Distress::high]/RefuseRequest <{0}

assentToSupportCalls: boolean
SupportCalled
CallSupportDecision()
PlatformOp
PlatformSupport
assent: boolean
ISupport
/assentToSupportCalls?assent <{0}
CallSupport
entry CallSupport(); support = true
[assent]
SupportCalled <{60}
[not assent]

assentToSupportCalls1: boolean
ctrl0
PlatformOp
PlatformSupport
IUserFall
IUserParam
IRoomTemp
IDressing
ICurtains
ISupportP
DressingStarted
roomTemperature: int
DressingAbandoned
DressingStarted
SupportCalled
ref stm_ref0 = DressingService
roomTemperature: int
DressingAbandoned
RetryAgreed
DressingComplete
assentToSupportCalls: boolean
RetryAgreed
DressingComplete

OpenCurtains()
/CurtainsOpened <{vt(60, Time::seconds)}
CurtainsOpened

Retry()
IRetry
/wait ([0,vt(159, Time::seconds)]); retry = true

Dress(temp: nat)
IRetry
[temp<17]/wait ([0,vt(60, Time::seconds)]); retry = true
[temp>=17/\temp<19]/wait ([0,vt(90, Time::seconds)]); retry = true
[temp>=19]/wait ([0,vt(2, Time::minutes)]); retry = true

CurtainOpenRqt
assentToSupportCalls: boolean
CurtainOpenRqt
SupportCalled
CurtainsOpened
userUnderDressed: boolean
CurtainsOpened
ref stm_ref1 = MonitoringService
RefuseRequest
userDistressed: Distress
UserFallen
userDistressed: Distress
RefuseRequest
UserFallen
Clear()
SupportCalled
CallSupportDecision()
OpenCurtains()
Retry()
Dress(temp: nat)
assentToSupportCalls: boolean
CurtainsOpened

Software

assentToSupportCalls2: boolean
assentToSupportCalls1: boolean
ctrl1
IAssents
assentToSupportCalls1: boolean
assentToSupportCalls2: boolean
stm0
assentToSupportCalls1: boolean
assent: boolean
IAssents
Relay
during assentToSupportCalls1!assent
assentToSupportCalls?assent
Relay2
during assentToSupportCalls2!assent
async
assentToSupportCalls: boolean
roomTemperature: int
SupportCalled
CurtainOpenRqt
CurtainsOpened
DressingAbandoned
RefuseRequest
DressingStarted
RetryAgreed
DressingComplete
DressingStarted
RetryAgreed
assentToSupportCalls?assent
assentToSupportCalls1: boolean
ctrl0
SupportCalled
PlatformSupport
supportCalled1
ref RAD::RADRobot
SupportCalled
supportCalled1
supportCalled2

RADRobot
assentToSupportCalls: boolean
PlatformOp
SupportCalled
IUserFall
IUserParam
IRoomTemp
IDressing
ICurtains
ISupport
PlatformSupport
userDistressed: Distress
DressingComplete
UserFallen
userUnderDressed: boolean
roomTemperature: int
DressingStarted
DressingAbandoned
RetryAgreed
CurtainOpenRqt
RefuseRequest
CurtainsOpened